THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS FULL SYLLABUS PAPER SET 06

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The number 1729 is famous as the: 1

(a) smallest number expressible as the sum of two cubes in two ways
(b) smallest perfect square
(c) sum of first 10 odd numbers
(d) largest perfect cube

2. The value of 25^2 is: 1

(a) 600 **(b)** 50 **(c)** 650 **(d)** 625

3. 7.5×10^4 written in usual form is: 1

(a) 75000 **(b)** 7500 **(c)** 750000 **(d)** 0.00075

4. The word 'algorithm' is named after: 1

(a) Al-Khwarizmi **(b)** Aryabhata **(c)** Euclid **(d)** Fibonacci

5. A rhombus is a parallelogram in which: 1

(a) one pair is parallel **(b)** all sides are equal
(c) all angles are 90° **(d)** diagonals are equal

6. The diagonals of a rectangle are: 1

(a) equal and bisect each other **(b)** perpendicular
(c) unequal **(d)** parallel

7. Is 2728 divisible by 11? 1

(a) No **(b)** Only if it ends in 1
(c) Only if it is odd **(d)** Yes

8. If $a = 4$ and $b = 9$, then $(a + b)^2$ equals: 1

(a) 133 **(b)** 25 **(c)** 169 **(d)** 97

9. The ratio 3 : 4 is equivalent to: 1

(a) 6 : 9 **(b)** 7 : 8 **(c)** 9 : 12 **(d)** 4 : 3

10. ₹ 660 is divided between A and B in the ratio 5 : 6. A's share is: 1

(a) ₹ 420 **(b)** ₹ 330 **(c)** ₹ 300 **(d)** ₹ 360

11. 12.5% of 480 is: 1

(a) 50 **(b)** 120 **(c)** 70 **(d)** 60

12. A quantity of 800 is increased by 25%. The new value is: 1

(a) 1200 **(b)** 200 **(c)** 600 **(d)** 1000

13. The hypotenuse of a right triangle is 17 cm and one leg is 8 cm. The other leg is: 1

(a) 9 cm **(b)** 15 cm **(c)** 16 cm **(d)** 14 cm

14. A ladder 10 m long reaches 8 m up a wall. The foot of the ladder is how far from the wall? 1

(a) 8 m **(b)** 4 m **(c)** 6 m **(d)** 2 m

15. 8 workers finish a job in 9 days. How many days will 12 workers take? 1

(a) 5 **(b)** 8 **(c)** 13.5 **(d)** 6

16. Which solid has a circular face and one vertex? 1

(a) cone **(b)** sphere **(c)** cylinder **(d)** cuboid

17. The range of 2, 3, 3, 5, 3, 7 is: 1

(a) 3 **(b)** 7 **(c)** 4 **(d)** 5

18. In a calendar, the dates in a 2×2 block starting at 31 are 31, 32, 38, 39. Their sum is: 1

(a) 132 **(b)** 142 **(c)** 140 **(d)** 138

19. The area of a parallelogram with base 9 cm and height 7 cm is: 1

(a) 32 cm² **(b)** 31 cm² **(c)** 63 cm² **(d)** 70 cm²

20. A square park of side 40 m has area: 1

(a) 400 m² **(b)** 800 m² **(c)** 160 m² **(d)** 1600 m²

SECTION — B (Very Short Answer Type) — 8 × 2 = 16

Answer in about 10–15 words each.

21. Write the number 3,500,000 in scientific notation. 2

22. Which digit appears in base 2 numbers? Write 5 in base 2. 2

23. Write the Kaprekar process for 3-digit numbers briefly. 2

24. If the ratio of ages of two sisters is 3 : 4 and the elder is 20, find the younger. 2

25. Write two triplets in which the smallest member is even and one with smallest member odd. 2

26. Find the missing term: 3 : 4 : 5 = 9 : x : y. 2

27. Is the mean always one of the data values? Give an example. 2

28. Find the area of a square of perimeter 52 cm. 2

SECTION — C (Short Answer Type) — 8 × 3 = 24

Answer in about 30–40 words each.

29. Is 1620 a perfect square? Use prime factorisation to decide. 3

30. The diagonals of a quadrilateral bisect each other. Which quadrilaterals can it be? Why? 3

31. Show by an example that $(a - b)^2 \neq a^2 - b^2$. 3

32. If 30% of a school's 1200 students play sports, how many do not? 3

33. State two uses of nets in everyday life. 3

34. Two numbers add to 22. Find the numbers that give the largest product and the product. 3

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35. Using $(n,\ \frac{n^2-1}{2},\ \frac{n^2+1}{2})$ find the triplet for $n = 7$ and verify. 3

OR

A ladder of length 41 m rests against a wall with its foot 9 m from the wall. How high up the wall does it reach? 3

36. A triangular field has base 30 m and height 8 m. Find its area and the cost of levelling at ₹ 12 per m². 3

OR

Find the area of a trapezium with parallel sides 16 cm and 24 cm and distance between them 15 cm. Also find the area of an equal-height rectangle with width 24 cm. 3

SECTION — D (Long Answer Type) — 4 × 5 = 20

Answer in about 80–100 words each.

37. (i) Evaluate $(5^0 + 4^{-1}) \times 2^2$. 5
(ii) Find x if $\left(\frac{3}{5}\right)^x \times \left(\frac{3}{5}\right)^{-4} = \left(\frac{3}{5}\right)^5$.

OR

(i) Explain why $2^{-3} = \frac{1}{8}$. 5
(ii) Simplify $\dfrac{2^{-3} \times 3^2}{6^{-1}}$.

38. (i) A shopkeeper sells two articles at ₹ 1200 each, one at 20% profit and the other at 20% loss. Find the overall gain or loss. 5
(ii) Explain.

OR

(i) Salary of ₹ 40000 is increased by 15% and then decreased by 10%. Find the final salary. 5
(ii) Net change percentage.

39. **Case study:** Five friends' pocket money (₹): 200, 250, 300, 350, 900. 5
(i) Find the mean.
(ii) Find the median.
(iii) Which better represents them?
(iv) What happens without 900?

OR

(i) A class of 30 has mean 62. A class of 20 has mean 72. Find the combined mean. 5
(ii) Compare.

40. (i) Derive the area of a trapezium by dividing it into two triangles. 5
(ii) Find the area of a trapezium with parallel sides 20 cm and 30 cm and height 12 cm.

OR

Case study: A farmer's field is a trapezium with parallel sides 60 m and 90 m, and the distance between them 40 m. 5
(i) Find the area.
(ii) Cost of ploughing at ₹ 2 per m².
(iii) The area in hectares (1 ha = 10000 m²).

END OF QUESTION PAPER — Total: 80 Marks

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